



DMFF: Deep multimodel feature fusion for building occupancy detection

Kailai Sun*

Department of the Built Environment, College of Design and Engineering, National University of Singapore, 117566, Singapore
Department of Automation, BNRist, Tsinghua University, Beijing, 100084, China

ARTICLE INFO

Keywords:

Building occupancy detection
Audio-image pair
Transformer
Multimodel fusion

ABSTRACT

The 2022 Global Status Report for Buildings and Construction (Buildings-GSR) indicates that construction activities have returned to pre-pandemic levels in most major economies, alongside more building energy consumption. To achieve the Net Zero emissions target by 2050, particularly in the post-pandemic era, accurate occupancy information is important to enhance building energy efficiency and improve occupancy comfort. While remarkable progress has been made in existing studies, they struggle to make full use of multi-sensor data to achieve high accuracy. Furthermore, there is a high expectation of the multimodel multi-temporal fusion by Transformer. In this study, we present a Transformer-based multimodal, multi-temporal feature fusion method (DMFF) for occupancy detection. To transfer domain knowledge from artificial intelligence into the building area, DMFF includes a pretrain-finetune pipeline and leverages pre-trained visual and sound models. Multiple pre-trained Transformer encoders are employed to extract features of different modalities. Then, we propose a self-attention mechanism for modality fusion to learn relationships among various sensors. Our DMFF method demonstrates superior performance on a real dataset, outperforming machine and deep learning methods (e.g., Convolutional Neural Networks, Random Forest, and Multilayer Perceptrons). Applied to a room setting, DMFF shows promising potential for building energy savings. The code and demo are accessible at https://github.com/kailaisun/multimodel_occupancy.

1. Introduction

According to the 2022 global status report for buildings and construction (Buildings-GSR) [1], the sector's operational CO₂ emissions and energy demand have reached an all-time high, exceeding the COVID-19 pre-pandemic peak of 2019 by 2% and 3%, respectively. Buildings-GSR shows construction activities have returned back to pre-pandemic levels, alongside more building energy consumption. In Fig. 1, the green line represents the baseline to achieve the zero-carbon building stock target by 2050. The blue line represents the real point of decarbonization achieved in the year. From Fig. 1, the buildings and construction industry is not on track to decarbonize by 2050. The point of decarbonization in 2021 is the pandemic outlier. After the COVID-19 Pandemic, there has been a negative rebound since 2020 in building decarbonization, with increased energy consumption and higher CO₂ emissions. The gap is about 9 points of decarbonization. In 2021 [2], the buildings sector represents 37% of global energy and CO₂ emissions and over 34% of energy demand globally. Besides, the buildings sector represents 40% of Europe's energy demand. Improving building energy efficiency has been the world's top priority in solving the energy crisis [3,4].

The Environmental Protection Agency (EPA) reports that more than three-quarters of the world's population live in urban regions and typically spend around 90% of their time inside buildings [5]. As urban dwellings continue to rise and building numbers grow, the need for comfortable indoor environments is becoming more pressing. Indoor environment quality (IEQ) is closely related to occupancy comfort (e.g., thermal, visual, indoor air and acoustic comfort), where thermal comfort has a great effect on occupancy satisfaction. The performance of building heating, ventilation, and air conditioning (HVAC) systems is usually determined by thermal comfort factors (e.g., temperature, humidity, clothing, and activity level). Besides, the World Green Building Council underscores that disruptive noise has decreased individuals' productivity and concentration by 66% [6]. Indoor acoustic comfort in buildings enables people to focus and stay productive. Recently, societal awareness of the impact of indoor thermal and acoustic comfort on individual well-being and health has been heightened [7]. In recent, society's awareness of the impact of indoor thermal and acoustic comfort on personal well-being and health has increased.

To achieve Net Zero emissions by 2050, especially in the post-pandemic period, it is important to reduce CO₂ emissions in buildings, with a particular emphasis on utilizing smart buildings (SB). Smart

* Correspondence to: Department of the Built Environment, College of Design and Engineering, National University of Singapore, 117566, Singapore.
E-mail address: skl23@nus.edu.sg.

List of Abbreviations

LDA	Linear Discriminant Analysis
AIGC	AI-Generated Content
ANN	Artificial Neural Network
CNN	Convolutional Neural Network
CO ₂	Carbon Dioxide
CPU	Central Processing Unit
DNN	Deep Neural Networks
DT	Decision Tree
FCU	Fan Coil Unit
GPU	Graphics Processing Unit
GRU	Gated Recurrent Unit
HMM	Hidden Markov Model
HVAC	Heating, Ventilation, and Air Conditioning
LSTM	Long Short-Term Memory
MLP	Multilayer Perceptron
OCC	Occupant-Centric Control
PIR	Passive Infrared sensor
ppm	Parts per million
RAM	Random Access Memory
RF	Random Forest
RH	Relative Humidity
RNN	Recurrent Neural Network
SVM	Support Vector Machine
TVOC	Total Volatile Organic Compounds
ViT	Vision Transformer
Wav2vec	Waveform to Vector
XGBoost	eXtreme Gradient Boosting

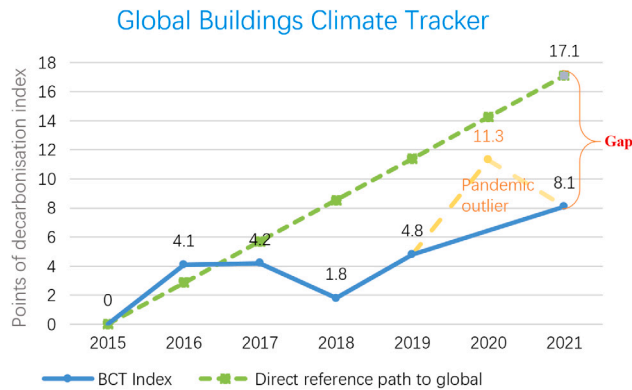


Fig. 1. Buildings Climate Tracker (BCT). There is a big gap between the expected decarbonization index and real operation [1].

buildings [8], which integrate advanced architecture, management, and automation technologies, aim to provide a suitable and efficient built environment [9]. Smart buildings employ automated systems to control actuators such as HVAC, lighting, and acoustics. SB systems can adjust environmental conditions, reducing energy usage and increasing comfort.

Occupancy information plays an important role in building energy efficiency, comfort improvement, building operation and retrofitting [10,11]. Building occupancy information can be divided into presence, counting [12], identity, tracking, behaviours, activities [13], and comfort (physiology and psychology). Considering the tight correlation between building energy consumption and occupancy, many studies have used occupancy information (e.g., presence and counting) to

enable occupant-centric control (OCC). Recent studies have verified that OCC can save 10%–40% building energy and improve thermal comfort by about 30% [14]. On the other hand, after the COVID-19 pandemic, occupancy information can help to control occupant flow and reduce disease transmission [15–17].

Existing studies often obtain occupancy information by sensors, including passive infrared (PIR) [18,19]), mmWave Radar, cameras, sound sensors, CO₂ sensors, WiFi, etc [20,21]. However, each sensor has different advantages and drawbacks [22]. In particular, PIR and mmWave Radar can quickly sense and control HVAC and lighting systems, but it is hard to handle static occupancy. CO₂ sensors can measure CO₂ concentration and detect occupancy information but suffer from the irregular opening of doors and windows. Sound sensors can record the sound intensity and detect occupancy but will be incorrect when occupants are quiet. Vision-based OCC methods (using cameras) often obtain fine-grain information, but they involve privacy concerns. With the rapid development of deep learning [23] and ChatGPT 4, state-of-the-art computer vision methods have been applied to occupancy detection. However, challenges include similar object disturbance, illumination, occlusion and cumulative errors.

To address these problems, existing studies use the multimodel fusion method to detect occupancy at the decision level. While existing studies have made remarkable developments, they struggle to make full use of sensor information and achieve high accuracy. Besides, there is a high expectation of the multimodel multi-temporary fusion by Transformer. We present DMFF, an end-to-end multimodel multi-temporary feature fusion method to detect occupancy information. DMFF adopts an advanced Transformer-based pretrain-finetune pipeline. We first introduce pre-trained ViT and Wav2vec2 for extracting image and sound features. Similarly, we use deep neural networks (DNN) to extract environment features. Then, we merge these different modality information and perform feature fusion with an attention mechanism. At last, we finetune the DMFF with ground-truth labels. To our knowledge, this is the first study to fuse multimodel and multi-sensor information at the feature level for building occupancy detection. The main contributions are as follows:

- We introduce the pretrain-finetune pipeline and pre-trained visual/sound models for building occupancy detection.
- To evaluate the performance, we provide an experimental comparison among our DMFF and different machine learning methods.
- We demonstrate that our DMFF achieves superior performance on a real dataset compared to other machine/deep learning methods (e.g., Convolutional Neural Network, Random Forest, Multilayer Perceptron). Our DMFF shows effective learning capabilities in multimodel occupancy detection.
- DMFF is applied to a room, demonstrating the potential in building energy saving. The code is available at https://github.com/kailaisun/multimodel_occupancy.

2. Related work

Accurately obtaining occupancy information is important in reducing energy consumption and carbon emissions. The occupancy information can be divided into occupancy presence, number, behaviours, identity, tracking, behaviours, etc. The occupancy detection task aims to detect the occupancy presence (binary information) of some people in a zone or not. The occupancy estimation/counting task aims to count/estimate the number of occupancy in a zone. The occupancy identity task aims to recognize the ID of every occupant. The occupancy tracking task aims to track the occupancy and obtain the moving trajectory.

Existing studies can be divided according to the sensor type, including PIR, camera, CO₂, smart meter, etc. Occupancy detection methods based on PIR are adept at swiftly managing HVAC and lighting systems,

Table 1
Related work.

Ref	Year	Sensor	Algorithm	Deep learning	Task	Fusion level
[24]	2024	Temperature, RH, CO2, TVOC	LSTM	✓	Estimation	Input
[25]	2024	Temperature, RH, light, smart plug	Markov Chain,XGBoost	□	Detection	Input
[26]	2024	PIR, temperature,light	Domain adaptation	✓	Estimation	Input
[27]	2023	CO2, pressure	ANN, RF, SVM	✓	Estimation	Input
[28]	2023	Temperature, RH, CO2	DT	□	Detection	Input
[29]	2023	Temperature, RH, FCU	Transformer	✓	Estimation	Input
[30]	2023	Temperature, CO2,smart meter	Deep clustering alignment, domain separation networks	✓	Estimation	Input
[19]	2023	PIR	GAN,CNN	✓	Estimation	Input
[31]	2023	CO2, sound	HMM	□	State forecasting	Input
[32]	2022	CO2, Wi-Fi	GRU	✓	Estimation	Input
[17]	2022	Camera	CNN	✓	Estimation	Decision
[33]	2022	Camera	LSTM	✓	Estimation	Decision
[34]	2022	WiFi ,camera	K-means, MLP	✓	Estimation	Decision
[35]	2022	Camera	CNN	✓	Estimation	Decision
[12]	2022	Camera	CNN	✓	Estimation	Decision
[9]	2022	Camera	CNN	✓	Estimation	Decision
[36]	2022	Camera,audio,CO2, temperature,RH	CNN,RF	✓	Detection	Decision
[37]	2021	CO2, temperature	MLP	✓	Detection	Input
[21]	2021	Camera	CNN	✓	Estimation	Decision
[38]	2021	Temperature	Rule	□	Detection	Input
[18]	2021	PIR	ANN	✓	Detection	Input
[39]	2021	WiFi	Poisson regression	□	Estimation	Decision
[40]	2020	Bluetooth	K-means	□	Detection	Input
[41]	2020	CO2	Correlation analysis	□	Estimation	Input
[13]	2020	Camera	CNN	✓	Activity	–
[42]	2020	Light, temperature, humidity, CO2	Particle swarm optimization, ANN	✓	Detection	Input
This study	2024	Camera, sound, CO2, RH, temperature,TVOC	Transformer	✓	Detection	Feature

yet they struggle with detecting stationary occupants. CO2 sensors, while capable of measuring CO2 levels and inferring occupancy, are less effective due to unpredictable door and window activities. Sound sensors can gauge occupancy through sound intensity but may yield inaccurate results in the absence of audible activity. Vision-based approaches using cameras provide detailed information but raise privacy issues. The advent of advanced deep learning techniques and the ChatGPT 4 model has led to their application in occupancy detection through state-of-the-art computer vision methods. Nevertheless, these approaches face hurdles such as interference from similar objects, variable lighting conditions, obstructions, and accumulating errors.

Besides, existing studies can also be divided into traditional methods and deep learning. We list the recent studies from 2020 to 2024 and summarize them in Table 1. In the fusion level, when authors feed their models with multi-sensor input or historical data, we mark this case as “Input”. When authors decide on the final occupancy information using with multimodel output, we mark this case as a “Decision”. In Table 1, a trend can be seen that many studies use multi-sensor (especially camera) fusion and deep learning methods to obtain occupancy information. This study is developed based on the Transformer and feature fusion.

3. Methods

This section offers detailed descriptions and insights into the core principles and working mechanisms of our methods. The method was established by two key parts: 1) The preprocessing of sensor data, including data normalization, data balance, etc. We first present the data balancing problem and multimode sensor data in existing real datasets. Then, we present our solutions for preprocessing data. 2) We introduce pre-trained deep learning models and apply the Transformer to the building occupancy detection area. The newly developed Transformer-based multimodel multi-temporary fusion algorithm is specifically designed for occupancy detection. Our DMFF network, used for occupancy detection, includes encoders, self-attention mechanism, and prediction head. We conduct a comparative analysis by evaluating the performance against machine/deep learning algorithms like CNN, RF, and MLP.

3.1. Data preprocessing

3.1.1. Data balancing

In practical buildings, due to the randomness of moving occupants, the occupied schedule is not stable. The occupied schedule is defined as

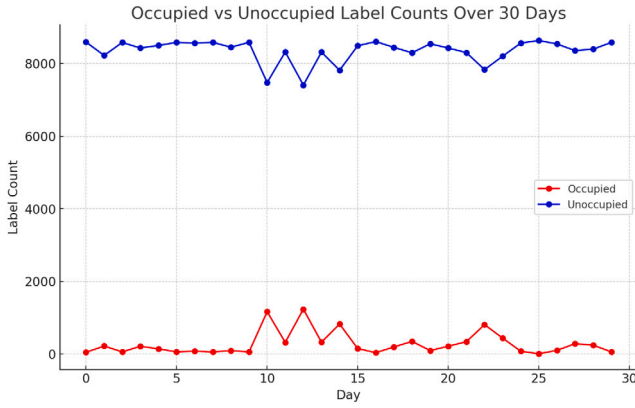


Fig. 2. Imbalanced building occupancy dataset. Source data from a real building [43].

the schedule when the room is occupied by occupants. The random occupied schedule brings some difficulties in deep learning applications. The common difficulty lies in the imbalanced dataset.

Imbalanced datasets can indeed present several challenges in deep learning, impacting the performance and accuracy of the models. When a dataset is imbalanced, meaning some classes are over-represented compared to others, the model tends to be biased towards the majority class. This can result in poor predictive performance and poor generalization of the minority class. Besides, deep learning models, with their high complexity, will overfit the majority class because we train them using more examples of the majority class. A model might achieve high accuracy by simply predicting the majority class most of the time, which does not reflect its true predictive power for the minority class.

An example is shown in Fig. 2. The figure illustrates the counts of ‘Occupied’ and ‘Unoccupied’ labels over approximately 30 days. The source dataset is a building occupancy dataset from the Ref. [43]. As observed, there is a significant imbalance between the two categories. The ‘Unoccupied’ label count is much higher than the ‘Occupied’ count on most days. This disparity can influence the performance of machine learning models, as they may become biased towards the more frequently occurring ‘Unoccupied’ label. Addressing this imbalance is crucial for developing effective and accurate predictive models.

To address this issue, we adopt an undersampling strategy for balancing the building occupancy dataset. We randomly sample sensor data with the ‘Unoccupied’ label in a room [43]. We get 12843 samples where the rate of ‘Unoccupied’ and ‘occupied’ labels is $\frac{6631}{12843-6631} \approx 1$. We use this balanced dataset for further analysis.

3.1.2. Multi-sensor data

To verify our proposed DMFF, we use multi-sensor data, including image data, environmental data (e.g., temperature, relative humidity, Carbon Dioxide Equivalence), and sound data. The detailed sensor information is in Table 2. We define $I(t)$, $S(t)$, $E(t)$ as image, sound, and environmental data, for a given timestamp t . It can be seen that different data modalities have different measurement ranges and data types. The raw data’s diverse distributions add complexity to the training process. To solve this, data normalization is necessary. This technique diminishes the risk of gradient explosion in deep learning, stabilizes training, and improves performance. Prior to algorithmic input, environmental data is standardized through data normalization:

$$E(t) = 2 * \frac{E(t) - E_{min}}{E_{max} - E_{min}} - 1, \quad (1)$$

where $E(t)$ is the environmental sensor value in the time t ; E_{min} denotes the minimum value; while E_{max} denotes the maximum value. Through this equation, all environmental data is converted to $[-1, 1]$.

In addition, because the image is different to the environmental data, for further input to pre-trained computer vision models, we resize the image to 224×224 and normalize it:

$$I(t) = \frac{I(t) - I_{mean}}{I_{sigma}}, \quad (2)$$

where $I(t)$ is the image in the time t ; I_{mean} denotes the mean value; while I_{sigma} denotes the standard deviation. Through this equation, the image is converted and follows the standard normal distribution.

As for the sound data, considering the existing pre-trained sound recognition models often need $[-1, 1]$ input, we first process an audio waveform by normalizing its values using Eq. (1). Then, we use the “librosa” library to interpolate and resample the waveform from 8 kHz to 16 kHz. This is because the pre-trained wav2vec2 model needs 16 kHz sound as its input.

3.2. Basic algorithms

For the convenience of understanding, we first introduce some basic machine learning algorithms we used in experiments. Then, we introduce deep learning algorithms and the Transformer.

3.2.1. Random Forest

Random Forest (RF) is a versatile ensemble learning technique suitable for both classification and regression tasks. This method constructs many decision trees during training and delivers the mode of the classes for classification tasks.

Delving deeper, RF enhances its predictive accuracy and mitigates overfitting through the creation of numerous decision trees, each based on a different sub-sample of the dataset and employing averaging. Specifically, each tree within the random forest sprouts from a bootstrap sample—that is, a randomly chosen subset of the original dataset, selected with replacement. The ensemble of these diverse trees then collectively contributes to a more robust and reliable prediction model. Bootstrap aggregating (bagging) is fundamental to RF’s approach to reducing overfitting.

During the training phase, RF selects a random subset of features for each tree. For classification problems, typically, the square root of the total number of features is chosen at random. This randomness injects diversity into the model, further enhancing its robustness. The final prediction made by a random forest model is an aggregate of the predictions made by each individual tree. The formula for this aggregation in RF can be expressed as:

$$f = \frac{1}{B} \sum_{b=1}^B f_b(x),$$

Where B is the number of decision trees, f_b is the prediction of the b th tree and x represents the input data for which the prediction is made. This collective decision-making process makes RF a powerful tool for predictive modelling.

3.2.2. Artificial Neural Networks

Artificial Neural Networks (ANNs) are sophisticated computational models inspired by the neural architecture of the human brain. They are a central technology in deep learning and machine learning. ANNs consist of a large number of neurons. Each neuron is capable of receiving, processing, and transmitting information. Within an ANN, the initial layer – the input layer – acquires raw data, which subsequently undergoes transformation across a series of hidden layers, each adding depth and complexity to the processing. At last, the output layer produces the final result. For each neuron, the computation follows:

$$y = f \left(\sum_{i=1}^n w_i \cdot x_i + b \right),$$

where y is the output of this neuron. f is the activation function, like Sigmoid, rectified linear unit (ReLU). w_i represents the weight

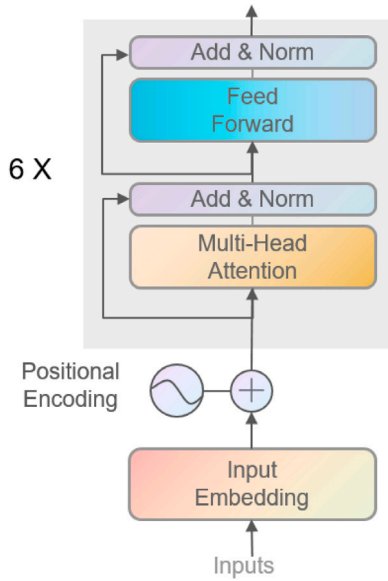


Fig. 3. An example of the Transformer encoder.

associated with the input. x_i is the i th input, which is multiplied by the weight w_i . b is the bias, which allows the output to be shifted.

ANNs can learn the patterns of a function by adjusting the weights. This learning typically occurs through a process known as backpropagation, where the network adjusts its weights by comparing its output to the desired outcome. This enables ANNs to learn complex patterns and relationships, making them highly effective in applications such as image and natural language processing.

3.2.3. Transformer encoder

With the rapid development of artificial intelligence (AI) and ChatGPT 4, the Transformer has been widely applied in different areas, especially in computer vision and natural language processing. We apply Transformer in building occupancy detection in this study. To achieve this, we introduce the basic Transformer encoder in this subsection.

Transformer encoder is a core component in machine/deep learning. The Transformer encoder is designed to process sequential input data (e.g., sound, video, text) by transforming it into a higher-dimensional space where relationships between different elements of the sequence can be learned more effectively.

At the base of the encoder in Fig. 3, we have the “Inputs”, which are typically tokens from a sentence that have been converted into vectors using an embedding process. These embeddings are then supplemented with “Positional Encoding”, which inserts information about the order of the tokens in the sequence.

After this, the “Multi-Head Attention” mechanism enhances the model to learn different parts of the input sequence simultaneously, enabling it to capture a wide array of dependencies and relationships within the data. The outputs of the attention mechanism are then normalized and passed through a “Feed Forward” neural network, which applies further transformations. The “Feed Forward” neural network usually includes two linear transformations and a non-linear activation function.

Each of these stages is followed by an “Add & Norm” step, which includes a residual connection. The residual connections can help to deal with the vanishing gradient problem by allowing gradients to flow through the network more easily, facilitating the training of deeper models.

3.3. DMFF

We apply the Transformer encoder for occupancy detection. It is suitable because the sensor data (e.g., image, sound, temperature, relative humidity) is sequential data. The Transformer encoder has the ability to handle parallel computations and capture complex relationships in data.

To achieve it, we present our DMFF in Fig. 4. Our DMFF can accept three different inputs: image, sound, and environmental data. To process them, we use pre-trained ViT and Wav2vec2 to extract image and sound features. Similarly, we use the Transformer encoder to extract environmental features. Then, we merge these different modality information and perform feature fusion with an attention mechanism. In summary, our DMFF includes three parts: encoder, self-attention mechanism, and prediction head.

3.3.1. Encoder

The sensor data contains different data modalities (e.g., image, sound, environmental data). An image is a matrix, and a video is a matrix sequence. A piece of sound data is a vector sequence. Temperature data is a scalar sequence. To handle and merge different data modalities, we first convert them to vectors.

To achieve it, we present a sound example in Fig. 4. The sound data (the left upper) is visualized by the time domain diagram. As we can see, the higher amplitude means a person performs actions in the room. To make it clear, we visualize the spectrogram in the left upper of Fig. 4. The interpretation is as follows: (1) activity around the 2-second mark: When the person appears and starts speaking/actions, we can see changes on the spectrogram plot. Human speech typically contains components across a range of frequencies. On the spectrogram plot, this might manifest as new bright spots or changes in colour (the left half of the plot). (2) after the person leaves: Once the person walks away and stops speaking, the spectrogram plot might show less activity or changes in colour, especially in the frequency areas common for speech. This indicates that the main component of the signal (i.e., the speech) is gone, possibly leaving behind some low-level background noise or silence. To capture these patterns, we use a pre-trained Wav2vec2 model [44]. The model is trained on the 960 h of Librispeech dataset on 16 kHz sampled speech audio. We transfer the knowledge in the Wav2vec2 model to the building occupancy area. It has two core encoders: (1) convolutional encoder: The Wav2vec2 model starts with a feature encoder that consists of several convolutional layers. This encoder processes raw audio waveforms and converts them into latent representations. These layers are designed to capture local features in the sound data. (2) Transformer encoder: The core of Wav2vec 2.0 is a context network based on the Transformer architecture. This network processes the latent representations and captures the contextual relationships within the sound data. The Transformer’s attention mechanism allows the model to focus on different parts of the sound sequence, making it adept at understanding longer-term dependencies.

For a given timestamp t and time interval Δt , we convert the sound sequence (16 kHz $\times$ 10 s = 160,000 points) to a sound vector (128 dimensions):

$$SV(t) = f_{\text{Wav2vec2}}(S(t), S(t+1) \dots S(t+\Delta t)). \quad (3)$$

Similarly, we present a video example in Fig. 4. In the left middle of the figure, we find that a person occurs in the room at the time $t+1$. Then, this person leaves the room in the following video. To capture this pattern, we use a pre-trained ViT model, which is trained on the ImageNet dataset (including 14 million images). We transfer the knowledge in the ViT model to the building occupancy area. First, the input image is split into many patches. Then, each patch is flattened into a one-dimensional vector and then transformed into a high-dimensional embedding using a trainable linear projection. Positional embeddings are added to keep the order information. Next, the sequence of embedded patches is processed through multiple Transformer encoder blocks.

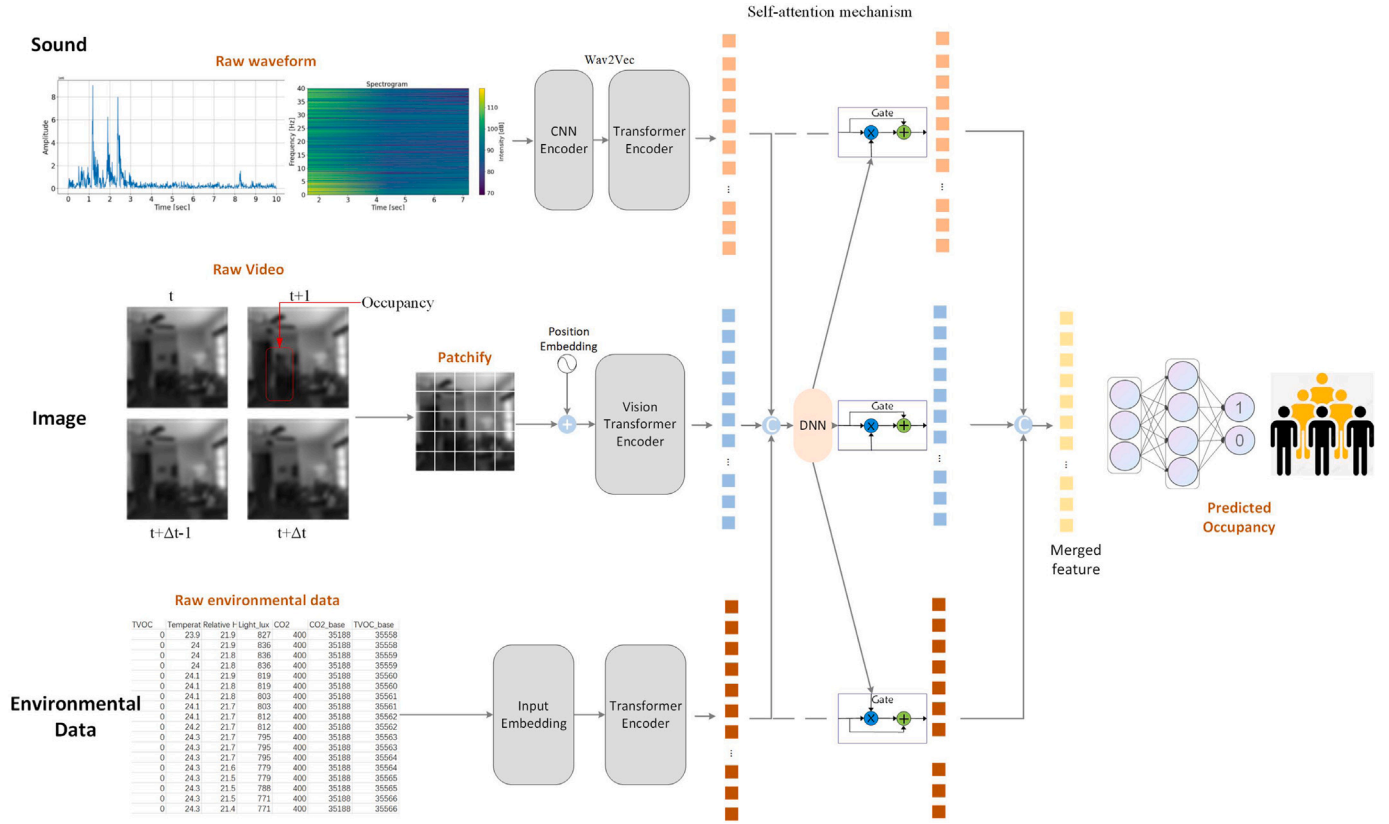


Fig. 4. Framework of DMFF.

Each block contains a multi-head self-attention mechanism, a Feed-Forward network, “ADD & Norm”, as shown in Section 3.2.3. Next, we average the multi-frame features to the same dimensions. In summary, for a given timestamp t and time interval Δt , we convert the image sequence ($224 \times 224 \times 3 \times 10 \text{ s} = 1,505,280 \text{ pixels}$) to an image vector (128 dimensions):

$$IV(t) = f_{\text{VIT}}(I(t), I(t+1) \dots I(t+\Delta t)). \quad (4)$$

Similarly, we present another example of environmental data processing in Fig. 4. In the bottom left of the figure, we find that the environmental data changes when a person is in the room. The temperature is increasing, and light is decreasing. To capture this pattern, after data normalization in Eq. (1), we add an input embedding to increase the dimension from 1 to 128, for each environmental data. Then, we apply a Transformer encoder with four heads in the multi-head attention models to extract the relationship between different sequences. In summary, for a given timestamp t and time interval Δt , we convert the environmental data sequence ($7 \times 6 = 42 \text{ points}$) to an environmental vector (128 dimensions):

$$EV(t) = f_{\text{Transformer}}(E(t), E(t+1) \dots E(t+\Delta t)). \quad (5)$$

3.3.2. Self-attention mechanism

Now, we have image-sound-environment vector pairs ($SV(t)$, $IV(t)$, $EV(t)$), for each given sensor data. Next, we merge these different modality information and perform modalities fusion with a proposed attention mechanism.

First, we concatenate the three different modalities:

$$V(t) = \text{Cat}(SV(t), IV(t), EV(t)), \quad (6)$$

where Cat means that the $V(t)$ is the concatenation of the three different modalities. Then, we propose a self-attention mechanism. We use the fully connected network to extract the relationship between the

three modalities. We separate the merged features into three different modalities:

$$V_1(t), V_2(t), V_3(t) = \text{DNN}(V(t)). \quad (7)$$

Next, for each modality, we use the Sigmoid activation function to construct a learnable mask (the range of the Sigmoid function is 0-1) and predict the score of the features so that more effective features can be selected:

$$SV'(t) = SV(t) \odot \text{Sigmod}(V_1(t)) + SV(t), \quad (8)$$

$$IV'(t) = IV(t) \odot \text{Sigmod}(V_2(t)) + IV(t), \quad (9)$$

$$EV'(t) = EV(t) \odot \text{Sigmod}(V_3(t)) + EV(t), \quad (10)$$

Last, we re-concatenate the modalities for further prediction:

$$V'(t) = \text{Cat}(SV'(t), IV'(t), EV'(t)). \quad (11)$$

3.3.3. Prediction head

Now, we have the feature $V'(t)$, which can represent the merged modality information. In Fig. 4, we use the standard DNN classifier to convert the feature $V'(t)$ to predicted occupancy. We use a 4-layer DNN architecture, a simple yet effective feedforward neural network.

The first layer is a fully connected (or dense) layer that reduces the dimensionality from $128 \times 3 = 384$ to 32. It performs a linear transformation on the input data. Following the first linear transformation, the second layer applies the ReLU6 activation function. (ReLU6 is similar to the standard ReLU but limits the maximum output value to 6. This can help prevent issues with large activation values in the network.) Then, the third layer further processes the data by transforming the 32 features from the previous layer to a number of features equal to 2. This transformation aligns with the number of classes in building occupancy prediction. At last, the final layer is a Softmax layer. This layer converts the output of the previous layer into a probability distribution over the 2 classes.

Table 2
Detailed sensor information.

Sensors	Units	Hardware	Range	Type
Camera	Pixel	Raspberry Pi Camera	(0,255)	Matrix
Air Temperature	Degree (°C)	Aosong DHT22	(−40,125)	Scalar
Relative Humidity	Percentage (% rH)	Aosong DHT22	(0, 100)	Scalar
Ambient light levels	Lux (lx)	Avago APDS-9301	N/A	scalar
Carbon Dioxide Equivalence	Parts per million (ppm)	Sensirion SGP30	(400, 60,000)	Scalar
Total Volatile Organic Compounds	Parts per billion (ppb)	Sensirion SGP30	(0,60,000)	Scalar
Sound sensor	N/A	Knowles SPH0645	N/A	Vector

4. Experiments and discussion

4.1. Performance evaluation

For the training phase, we used the cross-entropy loss as our objective function:

$$l(\theta) = - \sum_{i=1}^C y_i \log(p_i), \quad (12)$$

where C is the number of classes; y_i is the class label; p_i is the predicted probability of the observation being of class i .

To detect occupancy, we define an evaluation indicator:

$$\text{Accuracy} = \frac{1}{T} \sum_{t=1}^T (1 - \text{sign}|y(t) - o(t)|), \quad (13)$$

where $y(t)$ denotes the actual occupancy in the time t , and $o(t)$ represents the detected occupancy. The total number of samples is denoted by T .

Similarly, we use two evaluation indicators to evaluate our proposed model:

$$\text{Recall Rate} = \frac{\sum_{t=1}^T \mathbb{1}[y(t) = 1 \wedge o(t) = 1]}{\sum_{t=1}^T \mathbb{1}[y(t) = 1]}, \quad (14)$$

$$\text{Precision Rate} = \frac{\sum_{t=1}^T \mathbb{1}[y(t) = 1 \wedge o(t) = 1]}{\sum_{t=1}^T \mathbb{1}[o(t) = 1]}, \quad (15)$$

where $\mathbb{1}[\cdot]$ represents the indicator function that equals 1 when its argument is true and 0 otherwise. $\mathbb{1}[y(t) = 1 \wedge o(t) = 1]$ represents that the actual occupancy is true and our model also predicts true. The recall rate measures the proportion of actual positive cases that our model correctly predicts, while the precision rate measures the proportion of positive identifications predicted by the model that is actually correct.

4.2. Hyperparameters

Since the pre-trained Wav2vec2 and ViT models have been trained in the open datasets: 960 h of Librispeech dataset on 16 kHz sampled speech audio, ImageNet dataset (14 million images, 21,843 classes). We just finetune our DMFF with a small change. To achieve it, the Adam optimizer, with a learning rate of 0.00001, was chosen for its adaptability and efficiency in training deep neural networks. The batch size is set to 4. We choose the 3407 as our random seed. Our model was trained for 30 epochs, during which we monitored the validation accuracy to save the state of the best-performing model.

Experiments are conducted on a computer with Ubuntu OS, an AMD EPYC 7662 CPU, 1 TB RAM, and an NVIDIA RTX 3090 GPU. We use Python3 and Pytorch as our programming platform and our deep learning library.

4.3. Dataset

To conduct this research, we use the public dataset from [43]. We use one month of sensor data from hub 1 for analysis. To address the imbalanced problem, we randomly sample sensor data with an ‘Unoccupied’ label. We get 12843 samples where the rate of ‘Unoccupied’

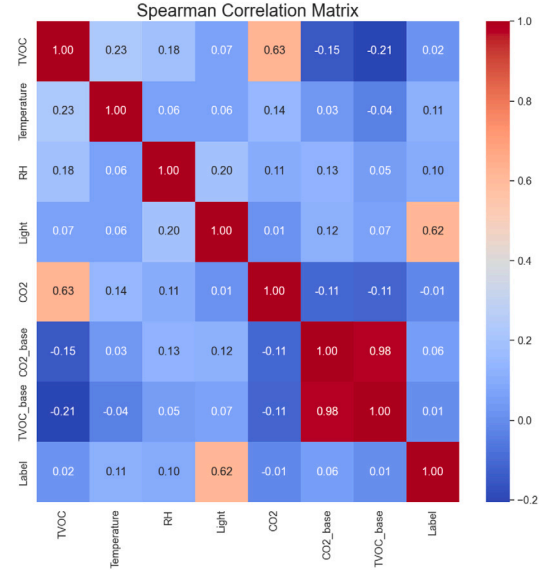


Fig. 5. Spearman Correlation Matrix.

and ‘occupied’ labels is $\frac{6631}{12843-6631} \approx 1$. We use this balanced dataset for further analysis.

This dataset has multi-sensor data, including image data, environmental data (e.g., temperature, relative humidity, Carbon Dioxide Equivalence), and sound data. The detailed sensor information is in Table 2. We use data normalization in Eqs. (1), (2). We split the dataset as the training set and validation set, with a ratio of 7:3.

4.4. Feature correlation

After obtaining the dataset, we conduct an analysis of feature correlation. Because the image and sound must have a strong correlation with occupancy. We analyse how environmental sensor data is related to occupancy. In Fig. 5, we conduct the Spearman correlation analysis. We find that the light level is more related to occupancy. It is reasonable because the occupants tend to open lights when coming into the room. The temperature and relative humidity (RH) are also related to the occupancy. It is reasonable because the occurrence of occupancy will change the environmental conditions, e.g., increasing the temperature and RH.

4.5. Performance comparison

After we finetune our DMFF model, we visualize the training process in Fig. 6. We can see that as the training epoch increases, loss decreases, and accuracy increases. The training and validation loss figure shows how the loss on both the training set and the validation set decreases over time. Loss is a measure of how well the model is performing; a lower loss means a better model. Training loss (blue line) shows a steady decrease, which indicates that the model is learning and becoming better at predicting the training data. The validation loss

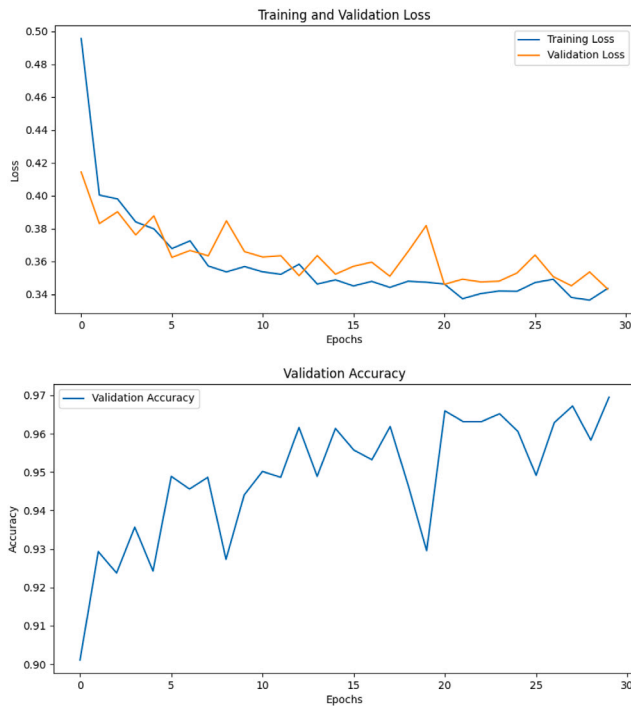


Fig. 6. Training process of DMFF.

(orange line) decreases along with the training loss but shows some upward spikes. This could indicate moments where the model does not generalize as well to the validation data. Ideally, both lines decrease and converge, indicating that the model is not overfitting (performing well only on the training data) or underfitting (performing poorly on both training and validation data).

The validation accuracy figure shows the accuracy of the DMFF model on our validation set, which is used to evaluate the model's performance. The high-accuracy beginning shows that the pre-trained models (Wac2vec2 and ViT) have a good performance on the building occupancy detection task. But it is not enough. The figure shows an overall increasing trend as the epochs progress. It indicates that the model is learning and improving its ability to correctly classify the validation data. There are fluctuations, which are normal during training, as the model adjusts its weights to minimize error. The goal is to have a validation accuracy that is high and stable.

In Fig. 7, we compare existing ML methods. The figure provides a clear visual comparison, demonstrating the relative performance of each model. The decision tree shows an accuracy of 90%. This model is generally considered to be more interpretable. We utilize 'gini' criteria and the unlimited maximum depth of the tree for training DT. MLP neural network: An accuracy of 88% is recorded for the Multi-Layer Perceptron, a type of neural network known for its ability to capture complex patterns through its multiple layers. The MLP model is configured with a maximum of 300 iterations, uses the 'ReLU' activation function, and employs the 'Adam' optimizer as the solver. With an accuracy of 89%, AdaBoost, a boosting type of algorithm that combines multiple weak learners to form a strong learner, performs slightly better than the MLP Neural Network. In the AdaBoost model, we set the maximum number of estimators is 50, and we use a better SAMME real boosting algorithm to improve performance. Logistic regression, often used for binary classification tasks, shows the lowest accuracy on the chart at 86%. In the LR model, we utilize the 'lbfgs' solver. The tolerance for stopping criteria is set as 0.0001. Linear Discriminant Analysis (LDA), which is particularly good for small datasets and when the data distribution is approximately normal, shows a high accuracy of 94%, tied with Random Forest for the second-highest accuracy. In the

Table 3

Confusion matrix. Positive means there are one or many occupants in the room. Negative means there is no occupant in the room.

	Predicted positive	Predicted negative
Actual positive	1947	35
Actual negative	76	1891

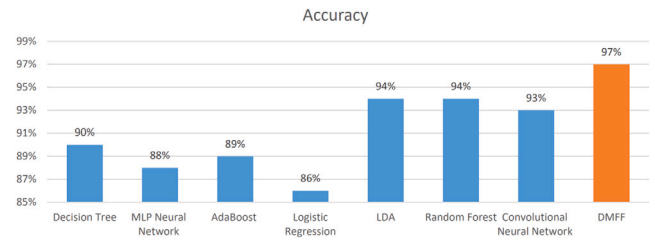


Fig. 7. Performance comparison.

LDA model, we utilize the Singular value decomposition solver which is recommended by the sklearn library for data with a large number of features, and we set the absolute threshold for a singular value of features as 0.0001. Random forest, an ensemble method that builds many decision trees and merges them together to get a more accurate prediction, also has a 94% accuracy. Like DT, we utilized 'gini' criteria and the unlimited maximum depth of the tree for training Random forest. Convolutional neural network is typically used for image processing. We use 1D CNN to detect occupancy, and this model shows an accuracy of 93%, slightly less than LDA and Random Forest. The CNN model implemented in PyTorch, featuring 3 convolutional layers with increasing channels (16, 32, 64), ReLU activations, max pooling, a flattening layer, 2 fully connected layers, and dropout for regularization. For training, the CNN model employs the binary cross-entropy with logits loss (BCEWithLogitsLoss), suitable for binary classification tasks, uses the Adam optimizer with a learning rate of 0.001, and is set to train for 50 epochs.

The DMFF model outperforms the others with the highest accuracy of 97%. In terms of relative performance, our DMFF model has a 3% higher accuracy than the next best models (LDA and Random Forest) and an 11% higher accuracy than the lowest-performing model (Logistic Regression). We draw the confusion matrix in Table 3. The recall rate is $\frac{1947}{1982} \approx 98.23\%$, and the precision rate is $\frac{1947}{2023} \approx 96.24\%$. In situations where errors occur, we observe that the DMFF's false negatives (35) are fewer than its false positives (76). This suggests that when controlling lighting or HVAC systems, a false negative, which indicates the presence of one or more occupants in the room, can lead the control system to erroneously switch off the lighting or HVAC. This results in a significant impact. Conversely, a false positive, implying the OCC system activates lighting or HVAC when there are no occupants, leads to only a minor increase in energy consumption. It can be accepted. In summary, the DMFF's performance indicates a significant improvement, suggesting that it might be better suited or better optimized for the building occupancy task at hand compared to the other models.

5. Conclusions

After the COVID-19 pandemic, energy conservation has increasingly become a global challenge. Accurate occupancy information is key to optimizing building energy use and enhancing environmental comfort. On the other hand, with the rapid advancements in AI and ChatGPT 4, some domain knowledge and pre-trained models can be transferred into the building area.

This study presents a Transformer-based method based on pre-trained computer vision and sound models. It is a Transformer-based

multimodal, multi-temporal feature fusion method (DMFF) for occupancy detection. We finetune our DFMM on a real indoor dataset in buildings. Our DMFF method demonstrates superior performance on a real dataset, outperforming machine and deep learning methods (e.g., Convolutional Neural Networks, Random Forest, and Multilayer Perceptrons). We apply our OCC strategy to the one-month dataset, demonstrating promising potential for building energy savings. To contribute to the efficient AI-driven approaches in energy conservation and building management, we make our code publicly available at https://github.com/kailaisun/multimodel_occupancy, fostering further research opportunities.

There is a limitation. We use the RGB image and sound data, which is related to privacy issues. In the future, we expect the researcher to think about achieving high accuracy while protecting privacy. For example, some studies based on lidar and infrared images are ongoing. On the other hand, most private occupancy datasets cannot be shared with others, but the trained models can be shared. These models are very useful. Some of them can be directly applied to other scenes. If these classification or regression models do not perform well, they can be seen as pre-trained models. Pre-trained models provide a good starting point for training specific custom models.

Another solution to prevent private issues is AI-generated content (AIGC). The potential benefits of AIGC are significant. By creating synthetic data, AIGC allows researchers to bypass many of the privacy issues associated with using sensitive or personal data. In fact, the synthetic data can be shared with global researchers. They can perform experiments, and train machine/deep learning models without risking the exposure of individual data points. By generating datasets that are free from privacy constraints, AIGC opens up possibilities for wider sharing of information. This can accelerate innovation and research without infringing on privacy.

CRedit authorship contribution statement

Kailai Sun: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

This research did not receive any specific grant from funding agencies in the public, commercial, or non-profit sectors.

Appendix

A.1. Energy saving

This study aims to develop an occupancy detection method that can be applied to save energy in buildings. This section uses a simplified example to demonstrate the potential of occupancy detection for energy saving.

We adopt an occupant-centric control (OCC) strategy to control equipment (e.g., air conditioning). We apply the OCC strategy to the one-month dataset (from 26 Nov to 26 Dec). The data was captured in a residential room in Boulder, Colorado, U.S. The minimum, mean, and

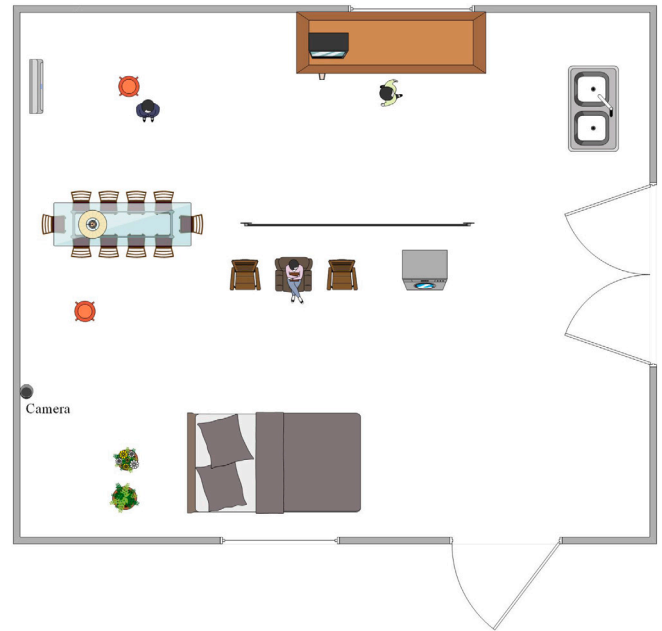


Fig. 8. House layout.

maximum outdoor temperatures in this area are -4.38°C , 2.06°C , and 11.22°C .¹ The house layout is shown in Fig. 8. A camera and other sensors are installed in the house. The house includes some furniture, such as tables, beds, sofas, and a kitchen. We use the real-world occupancy information from this dataset as input to control AC systems. Then, we compute the energy consumption using a basic strategy based on simulation. Next, we compute the energy consumption using our OCC strategy based on the same conditions (including temperatures, RH, etc.). We compare them and compute the energy-saving rate, keeping the time periods the same. Thus, the results of the two strategies are comparable.

In Fig. 9, we visualize an example of occupancy presence and effect of the OCC strategy. The room has four occupants. It can be seen that the occupants often enter and leave the room randomly. Due to the unknown occupancy status, to meet comfort needs, the building administrator usually uses the basic air condition control strategy (always open on working time). OCC strategy often controls equipment based on occupancy status. In this case, the occupants often forget to close the equipment when they leave. Thus, some energy can be saved with the OCC strategy. To conserve energy in buildings, it is crucial to minimize the operation time of the AC system during periods when the room is unoccupied.

To make it more clear, the energy saving rate R can be calculated as:

$$R = \left(1 - \frac{E_{\text{OCC}}}{E_{\text{Basic}}}\right) \times 100\% \quad (16)$$

- E_{Basic} is the energy consumption under the basic strategy.
- E_{OCC} is the energy consumption under the OCC strategy.

In Eq. (13), $o(t)$ represents the occupancy status function at time t , where $o(t) = 1$ indicates the presence of occupancy, and $o(t) = 0$ indicates no occupancy. Let $AC_{\text{basic}}(t)$ denote the air conditioning status function at time t , with $AC_{\text{basic}}(t) = 1$ indicating that the air conditioning is on, and $AC_{\text{basic}}(t) = 0$ indicating it is off. Let t_{start} be the start

¹ <https://psl.noaa.gov/boulder/Boulder.mm.html>

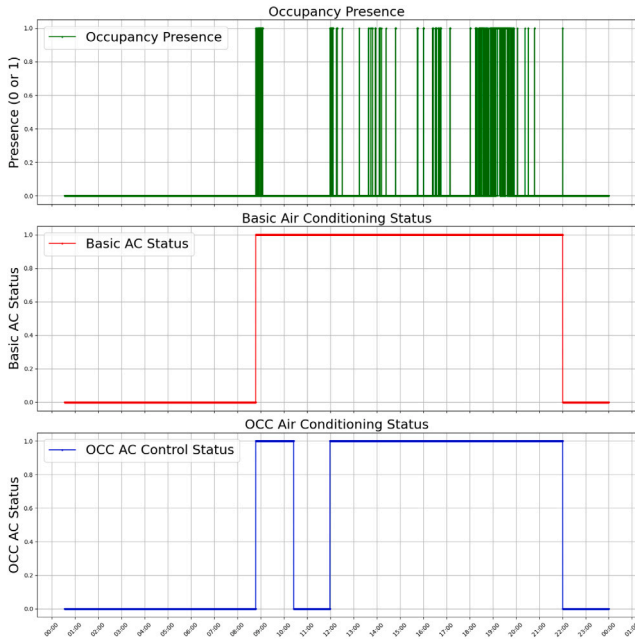


Fig. 9. An example of the effect of occupant-centric control.

time of air conditioning each day, and t_{end} be the time after the last person leaves. The air conditioning status $AC_{\text{basic}}(t)$ can be defined as:

$$AC_{\text{basic}}(t) = \begin{cases} 1 & \text{if } t_{\text{start}} \leq t \leq t_{\text{end}} \\ 0 & \text{otherwise,} \end{cases}$$

where t_{start} is the time when $o(t)$ first transitions from 0 to 1 within the day, and t_{end} is the time after the last occupancy detected. In mathematical terms, for each day, t_{end} can be calculated as:

$$t_{\text{end}} = \max(\{t \mid o(t) = 1\}).$$

For a set of discrete time points $t_1, t_2, \dots, t_n$, find the maximum time point t_i satisfying $o(t_i) = 1$ to determine t_{end} . Once t_{start} and t_{end} are determined, $AC(t)$ can be computed for each time point t to obtain a full daily sequence of the air conditioning status.

The air conditioning (AC) control strategy can be improved using our OCC strategy. The OCC strategy can be formalized as:

$$AC_{\text{OCC}}(t) = \begin{cases} 1 & \text{if } t_{\text{begin}} \leq t \leq t_{\text{last}} + D, \\ 0 & \text{otherwise,} \end{cases}$$

where t_{last} is the time of the last occupancy detected. The delay period D is added to t_{last} to compute the time until the AC is turned off. “ D ” means that when occupants leave the room, the AC cannot close immediately. Otherwise, it will reduce the useful life of the AC equipment. t_{begin} is the time of the beginning occupancy detected. To calculate t_{last} , we identify the latest time point t for which $o(t) = 1$. The AC is then set to remain on until $t_{\text{last}} + D$, after which it is turned off until the start of the next occupancy is detected. Following this, in Fig. 9, it can be seen that during 10:30–12:00, there is no occupant in the room, and our OCC strategy can help to close the AC and save energy.

We apply the OCC strategy in the one-month dataset (30 days). In modern buildings, a significant amount of electricity is wasted because equipment continues to operate even when no one occupies the room. Essentially, if we can accurately determine that a building or a room has been unoccupied for a specified period of time, we can save energy by turning off or reducing the use of certain facilities, like lights and air conditioning. Setting a lower threshold for this “unoccupied” time can lead to substantial energy savings. However, there is a trade-off: a lower threshold might result in frequent on-off cycles of the building’s

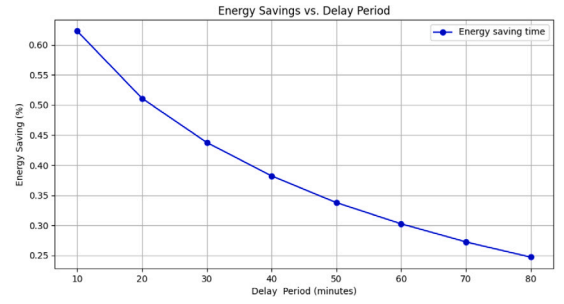


Fig. 10. Energy saving rate against delay periods.

equipment, which could reduce the useful life of the equipment. To provide a complete analysis, we set different delay periods and compare the energy saving rates in Fig. 10. The OCC strategy is that when an occupant enters the room, we open the equipment; when no one occupant is in the room, we close the equipment after a delay period. In Fig. 10, it can be seen that with the increasing control period, the energy saving rate is decreasing. If the control period is one hour, the OCC strategy can save about 30% of building energy. OCC strategy has a significant energy conservation potential.

References

- [1] U.N. Environment Programme, “2022 Global status report for buildings and construction”, 2022, <https://globalabc.org/our-work/tracking-progress-global-status-report>.
- [2] U.N. Environment Programme, “2021 Global status report for buildings and construction”, 2021, <https://globalabc.org/resources/publications/2021-global-status-report-buildings-and-construction>.
- [3] M. Economidou, V. Todeschi, P. Bertoldi, D. D’Agostino, P. Zangheri, L. Castellazzi, Review of 50 years of EU energy efficiency policies for buildings, *Energy Build.* 225 (2020) 110322.
- [4] Z.G. Ma, B.N. Jørgensen, Energy informatics: Tackling energy poverty in vulnerable communities, *Energy Inform.* (2023).
- [5] U. EPA, Buildings and their impact on the environment: A statistical summary, 2009, US Environmental Protection Agency Green Building Workgroup.
- [6] Z. Han, J. Kang, Q. Meng, Effect of sound sequence on soundscape emotions, *Appl. Acoust.* 207 (2023) 109371, <http://dx.doi.org/10.1016/j.apacoust.2023.109371>, URL: <https://www.sciencedirect.com/science/article/pii/S0003682X2300169X>.
- [7] L. Jiang, A. Bristow, J. Kang, F. Aletta, R. Thomas, H. Notley, A. Thomas, J. Nellthorp, Ten questions concerning soundscape valuation, *Build. Environ.* 219 (2022) 109231, <http://dx.doi.org/10.1016/j.buildenv.2022.109231>, URL: <https://www.sciencedirect.com/science/article/pii/S036013232200467X>.
- [8] S.S. Sameon, S. Yusoff, A. Abu Bakar, B.N. Jørgensen, Incorporating resilience into the IoT-based smart buildings architecture, in: B.N.r. Jørgensen, L.C.P. da Silva, Z. Ma (Eds.), *Energy Informatics*, Springer Nature Switzerland, Cham, 2024, pp. 202–212.
- [9] T. Xing, H. Yan, K. Sun, Y. Wang, X. Wang, Q. Zhao, Honeycomb: An open-source distributed system for smart buildings, *Patterns* 3 (11) (2022).
- [10] X. Jia, Y. Pan, M. Zhu, H. Zhu, Z. Li, J. Zhang, X. Zhou, S. Pan, C. Wang, D. Yan, et al., Occupant behavior modules development for coupled simulation in DeST 3.0, *Energy Build.* 297 (2023) 113437.
- [11] B. Dong, R. Markovic, S. Carlucci, Y. Liu, A. Wagner, A. Liguori, C. van Treeck, D. Oleynikov, E. Azar, G. Fajilla, et al., A guideline to document occupant behavior models for advanced building controls, *Build. Environ.* 219 (2022) 110386.
- [12] Y. Yang, Y. Yuan, T. Pan, X. Zang, G. Liu, A framework for occupancy prediction based on image information fusion and machine learning, *Build. Environ.* 207 (2022) 108524, <http://dx.doi.org/10.1016/j.buildenv.2021.108524>, URL: <https://www.sciencedirect.com/science/article/pii/S0360132321009197>.
- [13] P.W. Tien, S. Wei, J.K. Calautit, J. Darkwa, C. Wood, A vision-based deep learning approach for the detection and prediction of occupancy heat emissions for demand-driven control solutions, *Energy Build.* 226 (2020) 110386.
- [14] D.A. Winkler, A. Yadav, C. Chitu, A.E. Cerpa, Office: Optimization framework for improved comfort & efficiency, in: 2020 19th ACM/IEEE International Conference on Information Processing in Sensor Networks, IPSN, IEEE, 2020, pp. 265–276.
- [15] S. Liu, Z. Deng, Transmission and infection risk of COVID-19 when people coughing in an elevator, *Build. Environ.* 238 (2023) 110343, <http://dx.doi.org/10.1016/j.buildenv.2023.110343>, URL: <https://www.sciencedirect.com/science/article/pii/S0360132323003700>.

- [16] Evaluation of different air distribution systems in a commercial airliner cabin in terms of comfort and COVID-19 infection risk, *Build. Environ.* 208, 108590, <http://dx.doi.org/10.1016/j.buildenv.2021.108590>, URL: <https://www.sciencedirect.com/science/article/pii/S0360132321009823>.
- [17] K. Sun, Q. Zhao, Z. Zhang, X. Hu, Indoor occupancy measurement by the fusion of motion detection and static estimation, *Energy Build.* 254 (2022) 111593.
- [18] Y. Jin, D. Yan, X. Zhang, J. An, M. Han, A data-driven model predictive control for lighting system based on historical occupancy in an office building: Methodology development, *Build. Simul.* 14 (2021) 219–235.
- [19] J. Yun, D. Kim, D.M. Kim, T. Song, J. Woo, GAN-based sensor data augmentation: Application for counting moving people and detecting directions using PIR sensors, *Eng. Appl. Artif. Intell.* 117 (2023) 105508.
- [20] H. Choi, C.Y. Um, K. Kang, H. Kim, T. Kim, Review of vision-based occupant information sensing systems for occupant-centric control, *Build. Environ.* 203 (2021) 108064.
- [21] H. Choi, C.Y. Um, K. Kang, H. Kim, T. Kim, Application of vision-based occupancy counting method using deep learning and performance analysis, *Energy Build.* 252 (2021) 111389.
- [22] K. Sun, Q. Zhao, J. Zou, A review of building occupancy measurement systems, *Energy Build.* 216 (2020) 109965.
- [23] T. Dai, S. Liu, J. Liu, N. Jiang, Q. Chen, Development of a new dynamic smagorinsky model by an artificial neural network for prediction of outdoor airflow and pollutant dispersion, *Build. Environ.* 243 (2023) 110624, <http://dx.doi.org/10.1016/j.buildenv.2023.110624>, URL: <https://www.sciencedirect.com/science/article/pii/S0360132323006510>.
- [24] C. Lu, Enhancing real-time nonintrusive occupancy estimation in buildings via knowledge fusion network, *Energy Build.* 303 (2024) 113812, <http://dx.doi.org/10.1016/j.enbuild.2023.113812>, URL: <https://www.sciencedirect.com/science/article/pii/S0378778823010423>.
- [25] Z. Pang, M. Guo, B. Smith-Cortez, Z. O'Neill, Z. Yang, M. Liu, B. Dong, Quantification of HVAC energy savings through occupancy presence sensors in an apartment setting: Field testing and inverse modeling approach, *Energy Build.* 302 (2024) 113752, <http://dx.doi.org/10.1016/j.enbuild.2023.113752>, URL: <https://www.sciencedirect.com/science/article/pii/S0378778823009829>.
- [26] J. Dridi, M. Amayri, N. Bouguila, Unsupervised domain adaptation without source data for estimating occupancy and recognizing activities in smart buildings, *Energy Build.* 303 (2024) 113808, <http://dx.doi.org/10.1016/j.enbuild.2023.113808>, URL: <https://www.sciencedirect.com/science/article/pii/S0378778823010381>.
- [27] J. Kim, A. Choi, H.J. Moon, J.W. Moon, M. Sung, Occupancy estimation using IoT sensors and machine learning: Incorporating ventilation system operating state and preprocessed differential pressure data, *Build. Environ.* 246 (2023) 110979, <http://dx.doi.org/10.1016/j.buildenv.2023.110979>, URL: <https://www.sciencedirect.com/science/article/pii/S0360132323010065>.
- [28] S. Yun, D. Licina, Investigation of indicators for personal exposure and occupancy in offices by using smart sensors, *Energy Build.* 298 (2023) 113539, <http://dx.doi.org/10.1016/j.enbuild.2023.113539>, URL: <https://www.sciencedirect.com/science/article/pii/S0378778823007697>.
- [29] K. Sun, I. Qaisar, M.A. Khan, T. Xing, Q. Zhao, Building occupancy number prediction: A transformer approach, *Build. Environ.* 244 (2023) 110807, <http://dx.doi.org/10.1016/j.buildenv.2023.110807>, URL: <https://www.sciencedirect.com/science/article/pii/S036013232300834X>.
- [30] J. Dridi, M. Amayri, N. Bouguila, Unsupervised domain adaptation with and without access to source data for estimating occupancy and recognizing activities in smart buildings, *Build. Environ.* 243 (2023) 110651, <http://dx.doi.org/10.1016/j.buildenv.2023.110651>, URL: <https://www.sciencedirect.com/science/article/pii/S0360132323006789>.
- [31] O.A. Kabbaj, L.-M. Péan, J.-B. Masson, B. Marhic, L. Delahoche, Occupancy states forecasting with a hidden Markov model for incomplete data, exploiting daily periodicity, *Energy Build.* 287 (2023) 112985, <http://dx.doi.org/10.1016/j.enbuild.2023.112985>, URL: <https://www.sciencedirect.com/science/article/pii/S0378778823002153>.
- [32] Z.D. Tekler, A. Chong, Occupancy prediction using deep learning approaches across multiple space types: A minimum sensing strategy, *Build. Environ.* 226 (2022) 109689, <http://dx.doi.org/10.1016/j.buildenv.2022.109689>, URL: <https://www.sciencedirect.com/science/article/pii/S0360132322009192>.
- [33] R.C. Navarro, A.R. Ruiz, F.J.V. Molina, M.J.S. Romero, J.D. Chaparro, D.V. Alises, J.C.L. Lopez, Indoor occupancy estimation for smart utilities: A novel approach based on depth sensors, *Build. Environ.* 222 (2022) 109406, <http://dx.doi.org/10.1016/j.buildenv.2022.109406>, URL: <https://www.sciencedirect.com/science/article/pii/S0360132322006370>.
- [34] N. Alishahi, M.M. Ouf, M. Nik-Bakht, Using WiFi connection counts and camera-based occupancy counts to estimate and predict building occupancy, *Energy Build.* 257 (2022) 111759, <http://dx.doi.org/10.1016/j.enbuild.2021.111759>, URL: <https://www.sciencedirect.com/science/article/pii/S0378778821010434>.
- [35] K. Sun, P. Liu, T. Xing, Q. Zhao, X. Wang, A fusion framework for vision-based indoor occupancy estimation, *Build. Environ.* 225 (2022) 109631.
- [36] S.Y. Tan, M. Jacoby, H. Saha, A. Florita, G. Henze, S. Sarkar, Multimodal sensor fusion framework for residential building occupancy detection, *Energy Build.* 258 (2022) 111828, <http://dx.doi.org/10.1016/j.enbuild.2021.111828>, URL: <https://www.sciencedirect.com/science/article/pii/S0378778821011129>.
- [37] S.I. Kampezdou, A.T. Ray, S. Duncan, M.G. Balchanos, D.N. Mavris, Real-time occupancy detection with physics-informed pattern-recognition machines based on limited CO₂ and temperature sensors, *Energy Build.* 242 (2021) 110863, <http://dx.doi.org/10.1016/j.enbuild.2021.110863>, URL: <https://www.sciencedirect.com/science/article/pii/S037877882100147X>.
- [38] D.N. Wagner, A. Mathur, B.E. Boor, Spatial seated occupancy detection in offices with a chair-based temperature sensor array, *Build. Environ.* 187 (2021) 107360, <http://dx.doi.org/10.1016/j.buildenv.2020.107360>, URL: <https://www.sciencedirect.com/science/article/pii/S0360132320307290>.
- [39] N. Alishahi, M. Nik-Bakht, M.M. Ouf, A framework to identify key occupancy indicators for optimizing building operation using WiFi connection count data, *Build. Environ.* 200 (2021) 107936, <http://dx.doi.org/10.1016/j.buildenv.2021.107936>, URL: <https://www.sciencedirect.com/science/article/pii/S0360132321003401>.
- [40] Z.D. Tekler, R. Low, B. Gunay, R.K. Andersen, L. Blessing, A scalable Bluetooth low energy approach to identify occupancy patterns and profiles in office spaces, *Build. Environ.* 171 (2020) 106681, <http://dx.doi.org/10.1016/j.buildenv.2020.106681>, URL: <https://www.sciencedirect.com/science/article/pii/S0360132320300391>.
- [41] A. Franco, F. Leccese, Measurement of CO₂ concentration for occupancy estimation in educational buildings with energy efficiency purposes, *J. Build. Eng.* 32 (2020) 101714, <http://dx.doi.org/10.1016/j.jobe.2020.101714>, URL: <https://www.sciencedirect.com/science/article/pii/S2352710219316912>.
- [42] N. Fatema, H. Malik, Data-driven occupancy detection hybrid model using particle swarm optimization based artificial neural network, in: *Metaheuristic and Evolutionary Computation: Algorithms and Applications*, Springer, 2020, pp. 283–297.
- [43] J. Margarete, S.Y. Tan, G. Henze, S. Soumik, A high-fidelity residential building occupancy detection dataset, *Sci. Data* 8 (1) (2021).
- [44] A. Baevski, Y. Zhou, A. Mohamed, M. Auli, wav2vec 2.0: A framework for self-supervised learning of speech representations, in: *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 12449–12460.